



Forest structure as a determinant of grouse brood occurrence – An analysis linking LiDAR data with presence/absence field data

M. Melin^{a,*}, L. Mehtätalo^b, J. Miettinen^c, S. Tossavainen^d, P. Packalen^a

^a University of Eastern Finland, School of Forest Sciences, P.O. Box 111, 80101 Joensuu, Finland

^b University of Eastern Finland, School of Computing, P.O. Box 111, 80101 Joensuu, Finland

^c Finnish Wildlife Agency, Ratatie 41, 91501 Muhos, Finland

^d Finnish Wildlife Agency, Vanamokatu 3D, 96500 Rovaniemi, Finland

ARTICLE INFO

Article history:

Received 1 June 2016

Received in revised form 6 September 2016

Accepted 7 September 2016

Available online 12 September 2016

Keywords:

Lidar

Grouse

Remote sensing

Habitat

Forest structure

Ecology

ABSTRACT

Forest-dwelling grouse, and especially their broods, are highly dependent on forest and vegetation structure. In countries with intense forest management, it follows that the quality of their habitats is directly affected by forestry operations. Therefore, we must know which structural features of forests define a good grouse habitat and how the abundance of these features is affected by the forestry operations. Nowadays, airborne lidar (light detection and ranging) is being frequently used for forest inventories and terrain mapping. These data is becoming more and more publicly available and it holds detailed information about the forests and the vegetation structure; a key component of wildlife habitats.

In this study, we integrated lidar data with grouse brood presence/absence data. Through GLMM modelling, we aimed (1) to identify the structural features of forests that mostly determine grouse brood occurrence and (2) to assess how they are affected by forest management. The three species assessed were capercaillie, hazel grouse and black grouse. Depending on species, the brood presence was positively (and significantly) affected by denser shrub layer cover, denser canopy cover, higher canopies, or all of these features. The results indicate that grouse broods are highly susceptible to changes in forest structure.

In countries like Finland, game management is almost always implemented in managed forests, which in the light of our results creates a need to integrate habitat management into forest management. Our results suggest that when managing grouse brood habitats, attention should be given to maintain both, protective canopy cover and a good understorey cover. It is fair to assume that the removal of these components will significantly decrease brood occurrence. Further, the study showed that a dataset (lidar) collected more for the purposes of forestry can also be used to study wildlife habitats occurring often in the same forests.

© 2016 Elsevier B.V. All rights reserved.

1. Introduction

It is generally accepted that in forest ecosystems species richness and the overall availability of niches are dependent on forest structure and composition (Noss, 1990). This situation is highly topical in heavily forested countries that also manage their forests intensively. Here, the success of game and wildlife management is directly affected by forest management. It follows that for the wildlife, the structure of their current and future habitats is determined by forest management. The question of what constitutes suitable structural features of wildlife habitats and how abundant they are in managed forests is consequently of high importance.

* Corresponding author at: Yliopistokatu 7, 80101 Joensuu, Finland.

E-mail address: markus.melin@uef.fi (M. Melin).

Finland hosts three forest-dwelling grouse species: the capercaillie (*Tetrao urogallus*), the black grouse (*Tetrao tetrix*) and the hazel grouse (*Bonasa bonasia*), whose habitat quality is heavily affected by forest and vegetation structure and thus by forestry operations. All of these species are highly valued game species in Finland and they are hunted every autumn. In addition, the capercaillie is known to have the characteristics of an umbrella and an indicator species: habitats that are suitable for capercaillie are also suitable for the other forest grouses, and vital capercaillie populations are thought to indicate the well-being of other forest dwelling species too (Pakkala et al., 2003). Grouse populations in Finland have suffered from substantial declines during the past decades, especially in southern parts of the country (Helle et al., 2003). Suspected causes of this have been linked to over-hunting and predation, but also forestry: the methods of forest

management and how it changes the vegetation structure have had impacts on game management (Helle and Helle, 1991; Kurki et al., 2000; Sirkiä et al., 2010). The alternative prey hypothesis suggests that changes in forest structure have led to strengthened populations of small rodents that in turn have allowed increases in the populations of small predators, which also hunt grouse species especially during times of low rodent populations (Henttonen, 1989). Another notable factor is that grouse species in general tend to suffer from high chick mortality rates. Most die before the age of two months and mortality rates as high as 89% have been reported (Lindén, 1981). During their first weeks, chicks are dependent on insect food and so the presence of bilberry is of great importance. It is at first an indirect food source because it hosts many of the insect species that chicks use for food. Later as the chicks grow they switch from insect food to plant food and bilberry now becomes a direct and still important source of food (fresh leaves and stems) (Storch, 1993; Wegge et al., 2005). Hatching often takes place in June and at the time temperature is a key component of determining survival as young chicks are very intolerant to cold temperatures (Lindén et al., 1984).

When considering habitat structure and changes in it, the chicks, in particular, seem to be a highly susceptible group because of their low survival rates during the first months of life. Ludwig et al. (2008) found higher reproductive outputs for black grouse in unditched forests and Sjöberg (1996) noted higher growth rates for chicks in unditched forests, which they linked to the fact that ditched forests had reduced field layer vegetation. For capercaillie, diversified age-structure has been suggested to be an attribute of a good brood habitat (Broome et al., 2014). In a study in the temperate forests of South Korea, Rhim et al. (2015) noted that chick mortality of hazel grouse was lower in areas with dense understorey cover and that the low mortality levels were also affected by overstorey cover. In Fennoscandia, Brittas and Willebrand (1991) noted black grouse nesting success to be higher in areas providing greater nesting cover while Huhta et al. (2016) reported grouse brood presence to be positively correlated with old coniferous mixed forests due to the potentially rich bilberry cover in these habitats (offering insect foods in August). The importance of forest structure and its role in wildlife-habitat relationships has been known for long (MacArthur and MacArthur, 1961). In the past, however, it was practically impossible to quantify this structure accurately. Today, data about vegetation structure are widely available as many countries map their terrain and measure their forests with airborne laser scanners using lidar technology.

The measurements made by a lidar device are measurements of the distance between the device and the target. The distance is obtained by emitting pulses of laser light, which can be thought of as time-stamped clumps of photons. As a pulse hits a target, the photons reflect back. The device recognizes the incoming photons and measures the time between the emission and the arrival of the reflection (echo). Next, it calculates the location where the

echo came from, which is possible because the speed of light, the direction in which the pulse was shot, and the location where the pulse was shot from are known. In the end, the device produces point cloud data in which the X-, Y-, and Z-coordinates of every received echo, as well as the intensity at which they were received, are known. Modern systems submit as many as 800,000 pulses per second, resulting in a dense point cloud of data (Fig. 1).

The data shown in Fig. 1 provide accurate 3D information about the structure of forests and vegetation which from a wildlife perspective is directly linked to abundance of food, shelter, or nesting spots. Because of this, the use of lidar data in ecology is becoming more and more common (Vierling et al., 2008). The use of lidar together with wildlife survey data was pioneered by Hill et al. (2004) who linked lidar-based descriptions of tree canopy to bird species occurrences. The past decade has seen the data used to describe the relationships of vegetation structure and habitat use for various species such as primates (Palminteri et al., 2012), squirrels (Flaherty et al., 2014) and bats (Froidevaux et al., 2016) while it has also been linked with GPS-tracking (Melin et al., 2013, 2016; Lone et al., 2014). Since birds are highly dependent on the structure of vegetation and the tree canopy, they have been famous target animals in lidar-based analyses (Goetz et al., 2010; Hill and Hinsley, 2015). Finally, and most relevantly to our current efforts, Graf et al. (2009) and Zellweger et al. (2013, 2016) used lidar data to assess and model grouse habitats. While Vauhkonen and Imponen (2016) tested the integration of lidar-based classifications of important habitats into forest planning. They selected lidar attributes that were to depict the habitat requirements of capercaillie and hazel grouse. For reviews on the use of lidar to assess wildlife habitats, see Davies and Asner (2014), Hill et al. (2014) or Müller and Vierling (2014).

In sum, vegetation structure is an important feature of grouse habitats and lidar data can characterize this structure in details. In this study, we integrate grouse brood occurrence data with lidar-based descriptions about the structure of the surrounding vegetation. First, we hypothesize that the areas used by broods have a different structure from locations without broods. Second, we aim to uncover structural features of vegetation that mostly define a good grouse brood habitat. Finally, and more importantly, the results are viewed in the context of current recommendations of game-friendly forest management and forestry operations to assess their effects on grouse habitat quality.

2. Methods

2.1. Study areas

To assess the role of forest structure in different landscapes, analyses were performed in five study areas (Fig. 2) with varying landscapes and varying numbers of observed grouse broods.

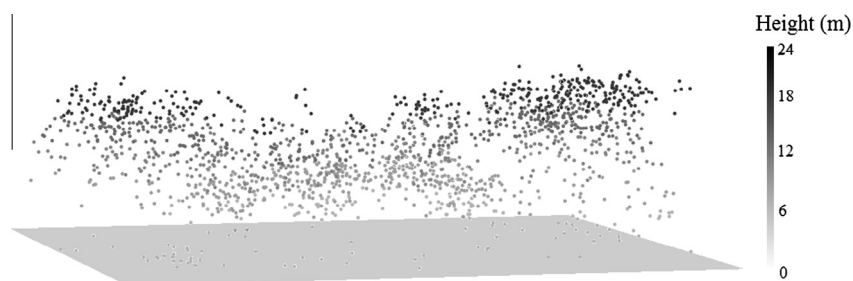


Fig. 1. Lidar data extracted and visualized from a 60 × 180 m-wide strip. Each of the echoes have XYZ coordinates, which allows us to quantify structural features such as canopy height and density, or amount of vegetation at different heights. The data in the image is so called discrete-return lidar data, where a return is counted and stored as an echo only if it is strong enough (has hit a big enough target).



Fig. 2. Location of the study areas (I–V).

The study areas are typical Finnish boreal forests with Scots pine (*Pinus sylvestris*), Norway spruce (*Picea abies*) and, to some extent, birch (*Betula* spp.) dominating the landscape. Additional deciduous species that may be present are European aspen (*Populus tremula*), grey alder (*Alnus incana*), rowan (*Sorbus*

aucuparia) and various *Salix* species. Study Area II has significantly more peatlands than the other areas. The northernmost, Study Area V, contains forests that are more Scots pine dominated and typically have fewer additional tree species (LUKE, 2015).

2.2. Grouse location data

The animal location data were provided by Natural Resources Institute Finland (LUKE). The used data were obtained from the wildlife triangle census, which is a long-term wildlife monitoring program. It dates back to 1988, when it was developed by the Finnish Game and Fisheries Research Institute (now LUKE) and the Hunters' Central Organization (now Finnish Wildlife Agency). The census provides annual information on the abundance of wildlife and the changes in its levels at the triangle locations. The triangle itself is an equilateral triangle whose sides measure 4 km. Altogether, there are over 1500 triangles located across Finland and around 1000 of these are surveyed annually. The triangle sides add up to form a 12 km long permanent inventory route. The route is surveyed by a three-person team (volunteers, local hunters), who walk the route and mark the locations of every animal observed along the way (inside a 60 m wide buffer). The route is checked twice a year (winter and summer) and the main purpose is to provide information on wildlife populations. (Lindén et al., 1996; Pellikka et al., 2005). The summer counting is done in July/August and its purpose is specially to estimate the grouse numbers and the success of grouse breeding prior to deciding the length and possible restrictions to the autumn's grouse hunting season. The data consists of observations made from grouse adults only (male or female, what species), broods, broods with adults, etc. With the late-summer triangle counts (see Helle and Lindström, 1991) the probability of detection is about 60% for single adults but close to 100% for broods (Brittas and Karlbom, 1990). Here, we utilized the summer survey data for the years 2014–2015 and focused only on locations with brood observations of three forest grouse species (capercaillie, black grouse, hazel grouse). Therefore, we felt safe to continue with the modelling given such high detection rates for grouse broods (Weidong and Swihart, 2004). The final data consisted of 98 wildlife triangles and 157 grouse brood locations (Table 1).

2.3. Data about the vegetation structure

Structure of vegetation was assessed from lidar data. The used data were part of a nationwide lidar dataset collected by the National Land Survey (NLS) of Finland and the Finnish Forest Centre. The data can be downloaded free from the NLS website (http://www.maanmittauslaitos.fi/en/file_download_service). The used data were collected between the years 2011 and 2014. The areas were scanned from an altitude of 2000 m above ground level using a field of view of 40 degrees. The nominal sampling density was about 0.8 measurements per square meter. The footprint of a single laser pulse was about 0.6 m at ground level. NLS classified ALS points as ground and non-ground points. In the processing phase, a digital terrain model (DTM) was interpolated from the ground points using inverse distance weighted interpolation (Shepard, 1968). Next, this DTM was subtracted from the orthometric heights of all the lidar echoes to scale their heights to above ground level (AGL) scale, providing every echo with an XYZ coordinate, in which

the Z coordinate stood for the AGL height. In general, lidar data collection and processing followed workflow that is presented in more detail in Melin (2015).

2.4. The study setting

The zone from which the grouse locations were observed in the field was 60 m nulls null wide, a 30-m buffer on both sides of the wildlife triangle inventory route line. For analytical purposes, this zone was divided into 60×60 m non-overlapping cells. This cell size was chosen, because the zone of observations is 60 m wide and also because the observation locations are not always marked down with a GPS. By using a 60×60 m, instead of e.g. 15×15 m, we have a higher probability that the observed brood really was where it was located at. Lidar data were then extracted from these cells. Fig. 3 illustrates the research setting. Corners of the triangles were omitted to avoid non-square or overlapping cells.

The lidar metrics were calculated cell by cell and thus they describe the structure of vegetation at the level of the 60×60 m cell. In addition, each cell was marked with by one or zero, indicating the respective presence or absence of a grouse brood. The three calculated lidar variables describe three important and unrelated features of the forest vegetation: amount of vegetation at the shrub layer (<5 m), canopy cover and canopy height (Tables 2 and 3).

Although features such as canopy height and cover are related (r^2 value in Table 3), they still can very efficiently differentiate forests: a young and dense forest may have higher canopy cover than a thinned mature forest, but the trend in height would be the opposite.

2.5. Modelling

The modelling task was to quantify the relationship between the presence of grouse broods and the lidar metrics. The response variable was binary with each cell codes as a '1' signifying the detection of a grouse brood or a '0' indicating no grouse broods detected. Also, the data were hierarchically grouped to plots (the 60 m cells) within clusters (the triangles) within the study areas. The study areas had differing forest and landscape structures from one another and so we could not treat the data as if the study areas were uniform. Therefore, we used generalized linear mixed-effects modelling (GLMM), which combines the properties of two statistical frameworks: mixed models (to account for the grouped structure) and generalized linear models (to handle non-normal data) (Bolker et al., 2008). The models were defined as:

$$\ln\left(\frac{p_{kji}}{1-p_{kji}}\right) = \beta_0 + \beta_1 x_{kji}^{(1)} + \beta_2 x_{kji}^{(2)} + \beta_3 x_{kji}^{(3)} + b_k + c_{kj}$$

where $\ln$ is a natural logarithm, p_{ijk} is the probability of brood occurrence in a cell i of triangle j in study area k and Y_{kji} are independent ($Y_{kji} \sim \text{Bernoulli}(p_{kji})$). Betas are coefficients of the three lidar variables x (Table 2). b_k is the random study area effect (that they are different from one another) and c_{kj} is the random triangle effect

Table 1
Summary of the grouse brood data.

Study area	Wildlife triangles	Brood observations by species			Total
		Capercaillie	Black grouse	Hazel grouse	
I	18	0	7	14	21
II	41	13	29	19	61
III	15	5	22	10	37
IV	8	5	3	4	12
V	16	11	13	2	26
Sum	98	34	74	49	157

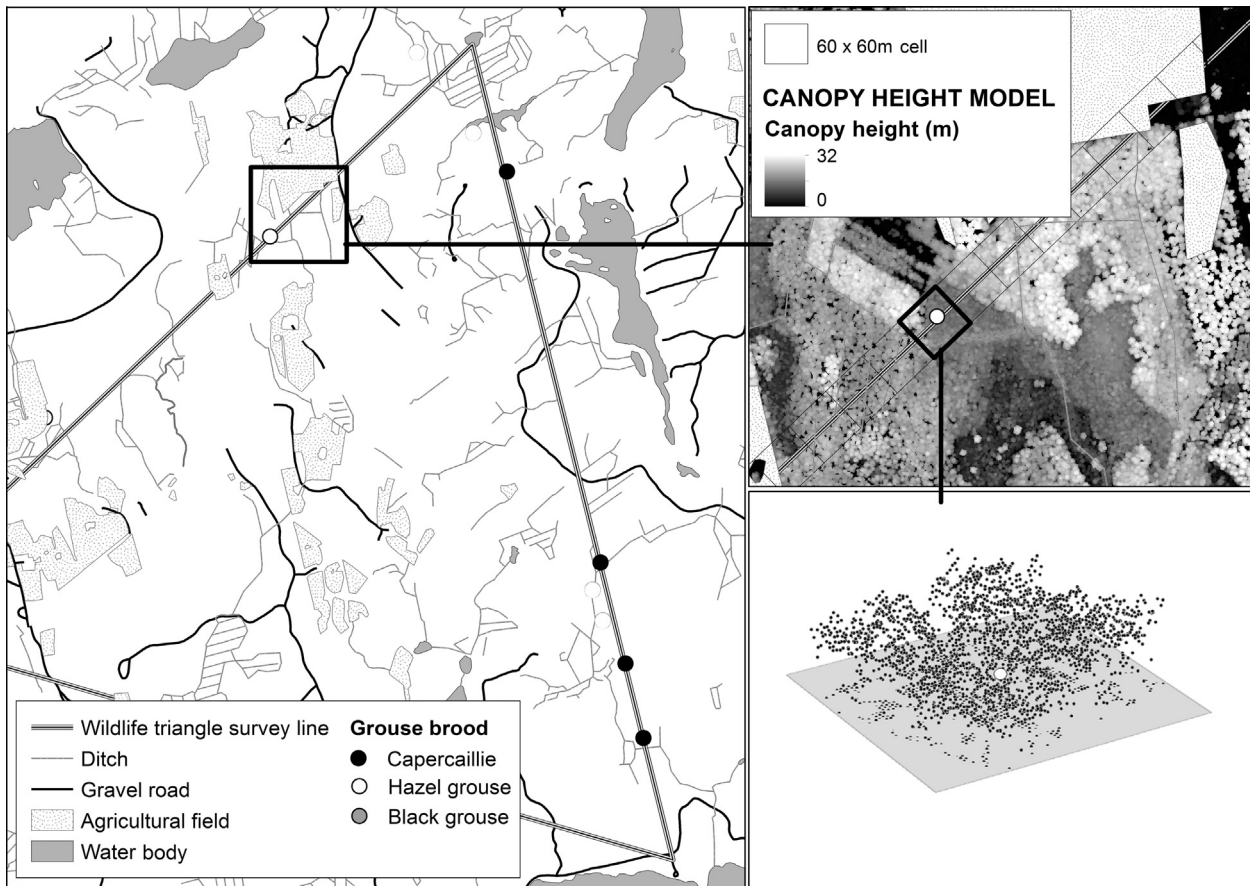


Fig. 3. The research setting.

Table 2

Variables calculated for each cell and descriptive statistics of their values in the data.

Variable	Description	Example	Range (min-max)	Mean	Standard deviation
Shrub_cover	Proportion of echoes between 0.5 and 5 m relative to all echoes reflected below five meters (%)	A <i>shrub_cover</i> value of 50 means that of all the echoes that reflected below five meters, 50% came from vegetation, not ground. The variable tells about the amount of vegetation between the heights of 0.5 and 5 m	0–91	20	14.88
Canopy_cover	Proportion of echoes above five meters (%)	A <i>canopy_cover</i> value of 70 means that 70% of all the echoes were reflected above five meters. The variable describes the amount of vegetation above 5 m of canopy cover: the higher the value, the denser the canopy (more vegetation)	0–98	38.55	24.37
Canopy_height	The 95th height percentile (meters)	A <i>canopy_height</i> value of 22.5 means that 95% of the echoes were reflected below the height of 22.5 m. The value of this variable indicates the maximum canopy height, which in the example case would be 22.5 m	0–31.1	15.02	5.41

Table 3

Correlations between the predictor variables.

r^2	Shrub_cover	cAnopy_cover	Canopy_height
Shrub_cover	1	0.25	–0.04
Canopy_cover	0.25	1	0.5
Canopy_height	–0.04	0.5	1

(triangles are nested within study areas) and $b_k \sim N(0, \delta_b^2)$ and $c_{kj} \sim N(0, \delta_c^2)$. Finally, via the inverse log-function, the predicted values were transformed to a scale from zero to one indicating the probability of brood occurrence. Models including all the variables listed in Table 2 were fitted for each of the three grouse species.

In Bernoulli data, variance and mean are related through

$$\text{var}(Y_{kji}) = p_{kji}(1 - p_{kji}).$$

However, often the data do not follow this due to lack of independence. It might show less or more variation than what would be expected based on a Bernoulli distribution. This is known as under- or over-dispersion, respectively. Over- or underdispersed Bernoulli data can be analyzed using Penalized Quasi Likelihood (PQL). In PQL, the GLMM is formulated as a nonlinear mixed-effects model, where the nonlinear function is determined by applied logit-link function and variance function by the above-mentioned theoretical variance, which is multiplied by a scaling factor σ^2 . The estimate of σ^2 indicates overdispersion if $\sigma^2 > 1$.

and underdispersion if $\sigma^2 < 1$. If potential overdispersion is ignored, we get smaller p-values (showing significance that might be false – we are anti-conservative) and potential underdispersion is ignored, we get higher p-values (showing insignificance even though the effect might be significant – we are conservative). Here, this issue was acknowledged and since the data was true presence-absence data, reliable inferences can be made from the reported p-values (Hastie and Fithian, 2013). We fitted our models using *glmmPQL* function from the MASS package (R Core Team, 2015).

3. Results

3.1. The effect of vegetation structure on grouse brood occurrence

The following tables show that there are significant differences related to vegetation structure between the areas where grouse broods are present or absent. There are also species-specific differences in terms of the variables (Table 2) which have the most significant effect on this. The tables show the models fitted for each of the species: capercaillie (Table 4), black grouse (Table 5) and hazel grouse (Table 6) broods. The fixed and random effect parameter estimates and the dispersion parameter in each of the tables come from the *glmmPQL* model fit.

Capercaillie broods were most affected by *canopy_cover* (Table 4, p-values). Because the parameter estimates are small, they can be directly interpreted as approximate relative changes in the probability of occurrence. For example, an increase of one percentage unit in *canopy_cover* or *shrub_cover* increases the probability of capercaillie brood occurrence by approximately 2% (Table 4). In practice, this suggests that the presence of capercaillie broods is positively correlated with increasing shrub and canopy cover. The species response to *canopy_height* was also positive, but insignificant in light of the p-values.

Table 4
The model for capercaillie broods.

Variable	Estimate	Std. error	p-Value
<i>Fixed effects</i>			
Intercept	−8.04	0.77	<2e−16
Shrub_cover	0.02	0.01	0.07
Canopy_cover	0.02	0.01	0.01
Canopy_height	0.03	0.03	0.44
<i>Random effects</i>			
Grouping	Std. dev.		
c_{kj} (triangle)	1.08		
b_k (study area)	1.15		
Under-/over-dispersion parameter			
σ^2	0.42		

Table 5
The model for black grouse broods.

Variable	Estimate	Std. error	p-Value
<i>Fixed effects</i>			
Intercept	−6.75	0.46	<2e−16
Shrub_cover	0.02	0.01	0.004
Canopy_cover	−0.01	0.004	0.06
Canopy_height	0.06	0.02	0.005
<i>Random effects</i>			
Grouping	Std. dev.		
c_{kj} (triangle)	1.03		
b_k (study area)	0.55		
Under-/over-dispersion parameter			
σ^2	0.56		

Table 6
The model for hazel grouse.

Variable	Estimate	Std. error	p-Value
<i>Fixed effects</i>			
Intercept	−6.92	0.40	<2e−16
Shrub_cover	0.02	0.01	0.004
Canopy_cover	0.02	0.004	0.0001
Canopy_height	−0.06	0.02	0.007
<i>Random effects</i>			
Grouping	Std. dev.		
c_{kj} (triangle)	1.62		
b_k (study area)	0.01		
Under-/over-dispersion parameter			
σ^2	0.34		

Black grouse broods showed significance to all three variables (Table 5). For black grouse broods, *canopy_cover* had a negative effect: as its value increases, the probability of black grouse brood occurrence decreases. For the other two variables, the trend was the opposite: *shrub_cover* and *canopy_height* were significant with positive estimates (Table 5). This suggests that increase in shrub cover increases the probability of black grouse brood occurrence, but for the canopy the species seems to prefer height over density, which is shown by the negative *canopy_cover*, but positive *canopy_height* effects. The most significant variable still was *shrub_cover*.

For hazel grouse broods, every variable was significant (Table 6). The effect of these variables for hazel grouse brood presence was positive for *shrub_cover* and *canopy_cover*, but negative for *canopy_height*. This implies that the species favors forests that have a dense shrub layer and dense canopy, but if the canopy gets too high, its effect becomes negative.

The dispersion parameter was less than one for all the species, which means that the data were under-dispersed (there was less variation than what might be expected). If the parameter value is one, the variation in the data is exactly as expected and if it is higher than one it is over-dispersed. Here, the values were 0.42 (capercaillie), 0.56 (black grouse) and 0.34 (hazel grouse), which indicate clear under-dispersion. If this had not been addressed under the PQL estimation, we would have obtained over-large p-values, which might have indicated insignificance instead of some of the noted significant effects shown in Tables 4 and 5.

3.2. Visualizing the structural differences

As the modelling revealed differences in vegetation structure between the brood and no-brood cells, we chose to visualize them. Fig. 4 uses histograms to illustrate these differences in brood and no-brood cells.

Fig. 4 illustrates neatly the trends shown in modelling. There are observable differences in the amounts of vegetation between the brood and no-brood locations. Most no-brood cells have shrub cover values between 0 and 20%, but in the brood cells these values are most commonly between 10 and 40%. We also see that the broods were in most cases absent from areas with the densest and highest canopies as well as from the areas with zero canopy cover or height.

4. Discussion

4.1. Points to note

In addition to the lidar-based analysis, we also tested a set of landscape metrics in the modelling. One of the metrics described

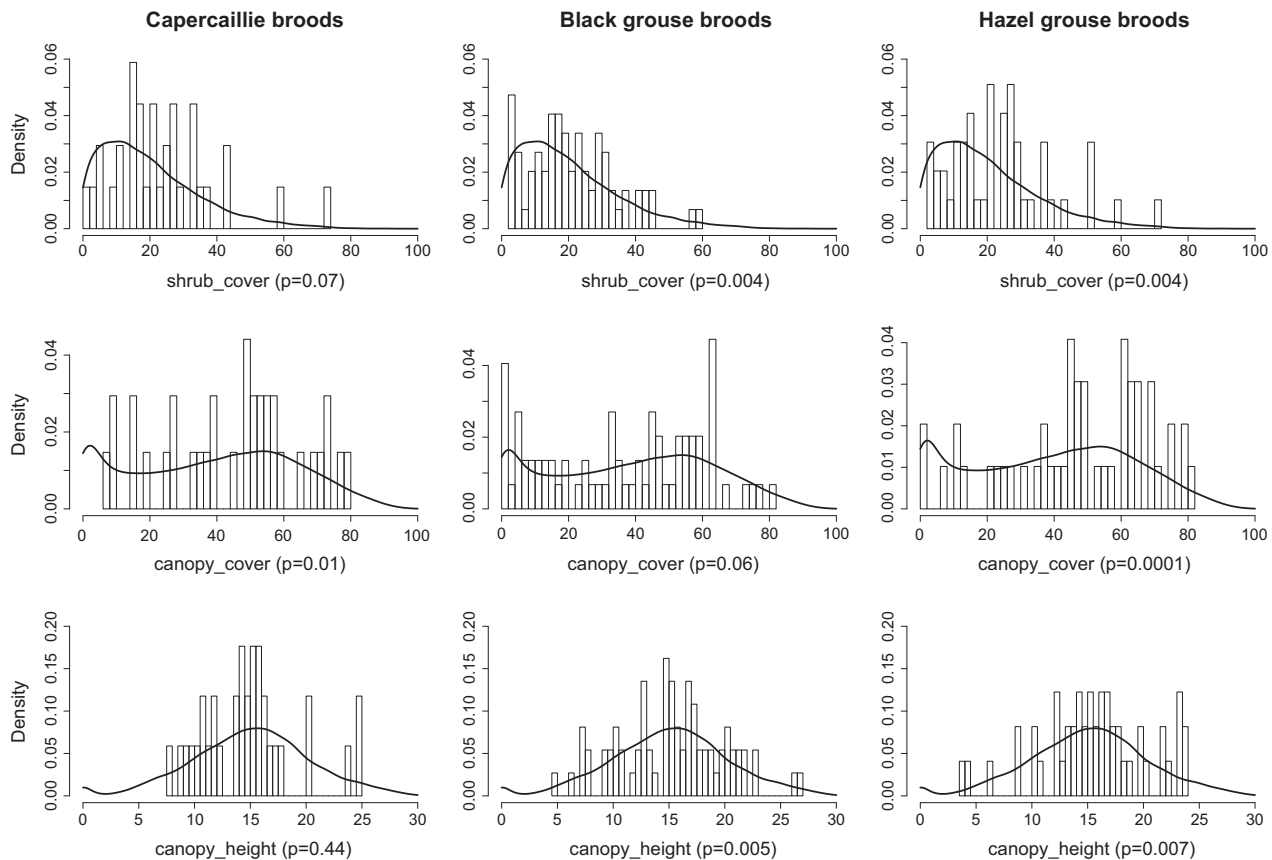


Fig. 4. Density histograms of the model variables from the brood cells. The bold line overlaying the bar histograms shows the density histogram of the same variable from the no-brood cells. Study areas are aggregated in these images. The p-values for each of the variables come from the modelling (Tables 4–6).

the amount of deciduous species in the cell (interpreted from Landsat images, available from Natural Resources Institute Finland: <http://kartta.mettla.fi/opendata/valinta.html>). The second set of metrics was the Euclidean distance from the center of each cell to the nearest water body, ditch, gravel road and agricultural field. These metrics gave direct information about the location of the cell in relation to the attributes. This data was downloaded from the same server as the lidar data. The data was polygon representations of the objects (like fields). In the end, none of these were significant whether they were used alone or in conjunction with the lidar variables. The situation would have surely been different if the view had been on, say, characterizing grouse habitat selection, which is known to occur on multiple scales (Lande et al., 2014; Miettinen et al., 2008; Kurki et al., 2000). Yet, as we were interested in the attributes that are affected by forest management, the landscape variables would not have been a target of most interest in this study. Lidar is evidently a highly useful tool in this study for assessing the importance of the vegetation structure. As the results showed, it did clearly differ between the areas where grouse broods were present and absent.

The reported predictors (Tables 4–6) were selected so that they are only weakly correlated to enable interpretation of the model parameters. Only the three reported lidar characteristics were evaluated as predictors to avoid bias in p-values (Harrell, 2001). Lidar data can yield various types of variables of the vertical profile of the vegetation. To select which ones to use, is always compromising between different option. The shrub layer could have been further delineated into two or more distinct layers (echoes between 0.5 and 2 m, between 2 and 4 m, etc.), but we felt safer to keep this threshold at between 0.5 and 5 m, because the main point, in the end, was to distinguish the canopy and shrub layers

from one another so that their structures could be analyzed separately. Also, if the limit had been lower (e.g. two meters), then the probability of catching the shrub layer with low-pulse density ALS would have been lower as well. With the limit of five meters, some pulses were certain to hit the shrub vegetation, if it existed in the amounts that would be meaningful as potential shelter. Certainly, if we had had lidar data with higher point density, we could have created variables that can describe the structure of the ground-floor vegetation in more details and also features like terrain ruggedness as well. Yet, as the data was low-pulse density discrete data, we chose not to create more detailed variables, because of the risk of getting more biased interpretations. The chosen variable (*shrub_cover*) can include lidar echoes that come from the lower branches of a big spruce or from a spruce seedling just as well. Still, it directly describes what it is supposed to: the amount of vegetation between 0.5 and 5 m. Furthermore, as the point was also in linking the results with forestry operations, the limit of five meters felt justified as well in relation to e.g. pre-commercial thinnings and to what kind of vegetation they erase. With the canopy layer above five meters, we did not make further vertical divisions (e.g. 5–10 m) and so anything above five meters was considered as the tree canopy. Canopy height variable was then informative, since it separated mature forests from younger ones (which could both have high canopy cover values).

4.2. What did the broods favor?

Depending on the species, the variable with the strongest effect on brood occurrence was either the amount of vegetation at the shrub level (*shrub_cover*) or the properties of the tree canopy above five meters (*canopy_cover* and *canopy_height*) or all of these.

The probability of capercaillie brood occurrence was increased by increased canopy cover, shrub cover, and canopy height (though insignificant), so that an increase of one percentage unit in canopy cover increases the probability by 2% ($p = 0.01$) and in shrub cover it increases it also by 2% ($p = 0.07$) (Table 4, Fig. 4). These findings agree with a general view that overall forest cover is a well-known characteristics of capercaillie habitats (Miettinen et al., 2010), but understorey cover is also a feature of good capercaillie habitats (Ludwig, 2007; Miettinen et al., 2010). Also, Fig. 4 shows that capercaillie broods were located in cells which had, on average, more shrub vegetation and canopy cover than no-brood cells. Ludwig (2007) noted the amount of horizontal cover was inversely associated with loss of grouse nests. Miettinen et al. (2010) then suggested that capercaillie need two types of cover: canopy cover (vertical direction) against avian predators and horizontal cover, close to the ground, against mammalian predators (see also Kvasnes and Storaas, 2007). Our results support these views that both of these variables (shrubs and canopy cover) are positively correlated with capercaillie brood occurrence (Table 4).

The probability of black grouse occurrence was positively correlated with *shrubs_cover* ($p = 0.004$) and *canopy_height* ($p = 0.005$). For instance, an increase of one percentage unit in *shrubs_cover* increased the occurrence probability by 2%. Interestingly, a similar increase in *canopy_cover* ($p = 0.06$) reduced this probability by 1% (Table 5). These findings agree with past studies suggesting that black grouse breeding as well as chick growth can be better in forests with more understorey cover (Picozzi and Hepburn, 1986; Brittas and Willebrand, 1991; Sjöberg, 1996; Ludwig et al., 2008). In this context, there were also indications that ideally this shrub vegetation should be located under a high canopy since the effect of *canopy_height* was also significant and positive (Table 5). The negative value of *canopy_cover* then suggests that regarding the canopy, height means more than density.

Hazel grouse brood occurrence was positively correlated with increasing *shrubs_cover* ($p = 0.004$) and *canopy_cover* ($p = 0.0001$) and negatively correlated with *canopy_height* ($p = 0.007$). An increase of one percentage unit in either *shrubs* or *canopy_cover* increased the probability of occurrence by 2% (Table 6). These results agree with Åberg et al. (2003), who found greater vertical cover (4–10 m) of vegetation as well as denser ground cover to be associated with hazel grouse occurrence. In addition, Swenson (1991) suggested that dense vertical vegetation cover is important for hazel grouse survival. Supposedly, dense canopy offers protection against goshawks (*Accipiter gentilis*) (Lindén and Wikman, 1983). In our model, *canopy_height* was negatively correlated with hazel grouse brood occurrence but positively with *canopy_cover*. Swenson and Angelstam (1993) found hazel grouse to use forests that are 20–50 years old and dense. Therefore it seems that as forests get too mature, their canopies are not dense enough, which is an attribute avoided also by hazel grouse broods. The age limit of 20–50 years (Swenson and Angelstam, 1993) and the vertical limit of 4–10 m (Swenson, 1991) suggest the same as our findings of an important canopy cover above five meters, but unimportant maximum canopy height. Hazel grouse broods showed interesting patterns in terms of the random effects (Table 6). The study area did not have any major effect on hazel grouse, but the variability between triangles was over twice as strong as with the other species. This implies that hazel grouse, as a species with the lowest dispersal ability, is therefore mostly affected by the local conditions (the triangles); variation was only shown within study areas, not between them as with the other grouse. Apparently, hazel grouse broods cannot utilize better habitats that might be located too far away in relation to their dispersal abilities.

Niemi et al. (1998) concluded that research can have an essential role in clarifying important ecological variables to sustain bird populations. This study showed that research based on lidar can

yield these variables for an important game and wildlife species: grouse. In the bigger picture, our results gave clear evidence that forest structure is an important attribute of grouse brood habitats. More specifically, increasing amounts of vegetation had a positive effect: the broods of capercaillie, black grouse and hazel grouse showed a clear preference for high shrub layer cover or high canopy cover or both. The findings support the views expressed in many previous studies across the world about the importance of understorey and overstorey cover for grouse broods (Picozzi and Hepburn, 1986; Brittas and Willebrand, 1991; Sjöberg, 1996; Broome et al., 2014; Ludwig et al., 2008; Jones et al., 2008; Rhim et al., 2015). However, the results gave also some new views on what defines a good habitat for each of the species and their broods. Here, instead of habitat separation (Swenson and Angelstam, 1993), our findings support the idea of relatively similar habitat preference among the broods of each species (in relation to forest structure). That is, despite having different ecological niches, the effect of forest structure and thus the potential effects of forest management are nevertheless significant.

4.3. Views on integrated forest- and grouse habitat management

In the ordinary commercially managed forests, canopy cover is abundant since it varies from moderate to high also after the early successional stages (Miettinen, 2009). In contrast, shrub layer cover may be completely absent. It is often removed by regular undergrowth pre-clearance, which is done to upgrade the efficiency of mechanical loggings in commercial harvestings. Our results showed that shrub cover is positively related to grouse brood occurrence and so its availability seems to be one key feature. Thus, if forest management methods are to be adjusted to respond to grouse habitat needs, attention should be given in maintaining both the protective canopy- and understorey cover. Given the results, it is fair to assume that the removal of these components will significantly decrease the probability of grouse brood occurrence. The data used in this study (lidar) is already used in forest inventory and planning purposes, but there are no doubts about its usability in wildlife research as well. Since much of boreal wildlife lives in managed forests, the data is valuable for both forest- and game managers, the latter of which in a country like Finland is directly dependent of the former.

Recently in Finland, the state-owned Tapio forestry consulting services published guidelines for integrated forest- and game management (Lindén et al., 2014) while Finnish Wildlife Agency published field guidelines for practices of game-friendly forest management (Finnish Wildlife Agency, 2015). These guidelines highlight, for instance, the importance of avoiding total clearance of the understorey vegetation and in keeping the vertical profile of the canopy diverse enough. These factors are emphasized during all the management operations: from tending of seedling stands and young forests to thinning's and possible pre-commercial clearances. Weight is also given to leaving small areas (e.g. 0.5–2 acres) with denser vegetation (*game thickets* in Finnish) outside the forestry operations throughout the operation chain, also during clear-cutting (3–5 thickets/hectare). These kinds of features have been noted as important in other countries as well (Finne et al., 2000; Hjeljord et al., 1995). In addition, managing the transition zones between different stands and sitetypes is mentioned as having particular importance especially from broods' point of view. Here, the retaining of proper understorey vegetation and vegetation with varying and diverse height distributions is, again, highlighted. When viewing these guidelines in the light of our results, the two agree very well indeed (Section 4.2). These operations, however, take place at the level of a single forest stand. For broods, who move within restricted areas, the effect is definitely instant. Yet, grouse habitat selection, as with many other species too, is

known to be a multi-scale approach (Lande et al., 2014; Miettinen et al., 2008; Kurki et al., 2000) with the good habitat being selected from a landscape that fulfills some of the basic requirements such as minimum amount of forest coverage. It is vital that this view is also taken in account when possible. Despite the importance, in Finland this is not always possible since most forests are owned by private people and the average size of a holding is only around 20–30 ha, meaning that this is also the unit size within which all the decisions on forest management are made. This, however, has been considered to some extent as advantageous, since it keeps the clear-cut areas rather small and thus prevents the formation of too large clear-cut areas (Lindén et al., 2014). Whatever the case, the mentioned guide lines are very good steps towards expanding game-friendly forest management.

When it comes to grouse habitats, future work should now focus on incorporating the other important variables along with vegetation structure (such as bilberry abundance) in the modelling of suitable grouse habitats. Here, a more detailed and accurate description of the ground-floor vegetation would be advantageous, but we were not able to perform it with the lidar data in hand. When it comes to integrating habitat- and forest management, the future work should focus on identifying important habitat features from lidar data and to integrate these into forest management and planning, which is more and more based on the same data. With grouse, the idea was already tested by Vauhkonen and Imponen (2016), who also suggested this to be a research topic for future. After all, forests will be managed in the future as well and so the mentioned integration would be the first steps towards efficient habitat management in commercial forests and forest planning throughout the boreal zone.

Acknowledgements

The corresponding author would like to acknowledge Suomen Riistanhoito- Säätiö (Finnish Game Management Foundation) for supporting this work.

References

- Aberg, J., Swenson, J.E., Angelstam, P., 2003. The habitat requirements of hazel grouse (*Bonasa bonasia*) in managed boreal forest and applicability of forest stand descriptions as a tool to identify suitable patches. *Forest. Ecol. Manage.* 175 (1–3), 437–444. [http://dx.doi.org/10.1016/S0378-1127\(02\)00144-5](http://dx.doi.org/10.1016/S0378-1127(02)00144-5).
- Bolker, B.M., Brooks, M.E., Clark, C.J., Geange, S.W., Poulsen, J.R., Stevens, H.H., White, J.-S.S., 2008. Generalized linear mixed models: a practical guide for ecology and evolution. *Trends Ecol. Evol.* 24 (3), 127–135. <http://dx.doi.org/10.1016/j.tree.2008.10.008>.
- Brittas, R., Willebrand, T., 1991. Nesting habitats and egg predation in Swedish black grouse. *Ornis Scandinavia* 14, 261–263.
- Brittas, R., Karlsson, M., 1990. A field evaluation of the Finnish 3-man chain: a method for estimating forest grouse numbers and habitat use. *Ornis Fennica* 67, 18–23.
- Broome, A., Connolly, T., Quine, C.P., 2014. An evaluation of thinning to improve habitat for capercaillie (*Tetrao urogallus*). *For. Ecol. Manage.* 314, 94–103.
- Davies, A.B., Asner, G.P., 2014. Advances in animal ecology from 3D-LiDAR ecosystem mapping. *Trends Ecol. Evol.* 29 (12), 681–691. <http://dx.doi.org/10.1016/j.tree.2014.10.005>.
- Finne, M.H., Wegge, P., Eliassen, S., Odden, M., 2000. Daytime roosting and habitat preference of capercaillie (*Tetrao urogallus*) males in spring – the importance of forest structure in relation to anti-predator behavior. *Wildlife Biol.* 6, 241–249.
- Finnish Wildlife Agency, 2015. Riistametsänhoidon työohjeet (In Finnish only) Available from: <http://riista.fi/wp-content/uploads/2015/09/Riistametsänhoidon-ty%C3%B6ohjeet-2015-Web.pdf>.
- Flaherty, S., Lurz, P., Patenaude, G., 2014. Use of LiDAR in the conservation management of the endangered red squirrel (*Sciurus vulgaris* L.). *J. Appl. Remote Sens.* 8, 1. <http://dx.doi.org/10.1117/1.JRS.8.083592>.
- Froidevaux, J.S.P., Zellweger, F., Bollmann, K., Jones, G., Obrist, M.K., 2016. From field surveys to LiDAR: shining a light on how bats respond to forest structure. *Remote Sens. Environ.* 175, 242–250.
- Goetz, S.J., Steinberg, D., Betts, M.G., Holmes, R.T., Doran, P.J., Dubayah, R., Hofton, M., 2010. Lidar remote sensing variables predict breeding habitat of a Neotropical migrant bird. *Ecology* 91 (6), 1569–1576.
- Graf, R.F., Mathys, L., Bollmann, K., 2009. Habitat assessment for forest dwelling species using LiDAR remote sensing: capercaillie in the Alps. *Forest. Ecol. Manage.* 257 (1), 160–167.
- Harrell, F.E., 2001. *Regression Modeling Strategies With Applications to Linear Models, Logistic Regression and Survival Analysis*. Springer, p. 568.
- Hastie, T., Fithian, W., 2013. Inference from presence-only data; the ongoing controversy. *Ecography* 36 (8), 864–867.
- Helle, P., Belkin, V., Bljudnik, L., Danilov, P.I., Jakimov, A., 2003. Changes in grouse populations in Finland and Russian Karelia during recent decades. *Suomen Riista* 49, 32–43 (In Finnish with English summary).
- Helle, P., Helle, T., 1991. How do changes in forest structure explain recent changes in Finnish grouse populations? *Suomen Riista* 37, 56–66 (In Finnish with English summary).
- Helle, P., Lindström, J., 1991. Censusing tetraonids by the Finnish wildlife triangle method: principles and some applications. *Ornis Fennica* 68, 148–157.
- Henttonen, H., 1989. Does an increase in the rodent and predator densities, resulting from modern forestry, contribute to the long-term decline in Finnish tetraonids. *Suomen Riista* 35, 83–90 (In Finnish with English summary).
- Hill, R., Hinsley, S., 2015. Airborne lidar for woodland habitat quality monitoring: exploring the significance of lidar data characteristics when modelling organism-habitat relationships. *Remote Sens.* 7 (4), 3446–3466.
- Hill, R.A., Hinsley, S.A., Broughton, R.K., 2014. Assessing habitats and organism-habitat relationships by airborne laser scanning. In: Maltamo, M., Næsset, E., Vauhkonen, J. (Eds.), *Forestry Applications of Airborne Laser Scanning: Concepts and Case Studies*. Springer, Netherlands, Dordrecht, The Netherlands, pp. 335–356.
- Hill, R.A., Hinsley, S.A., Gaveau, D.L.A., Bellamy, P.E., 2004. Predicting habitat quality for great tits (*Parus major*) with airborne laser scanning data. *Int. J. Remote Sens.* 25 (22). <http://dx.doi.org/10.1080/0143116031000139962>.
- Hjeljord, O., Spidsö, T.K., Björmyr, F., Meisingset, E., Dokk, J.G., 1995. Selection of birch by Black grouse (*Tetrao tetrix*) in winter. In: *Proceedings of the 6th International Grouse Symposium*, Udine, Italy, pp. 63–66.
- Huhta, E., Helle, P., Nivala, V., Nikula, A., 2016. The effect of human-modified landscape structure on forest grouse broods in two landscape types. In: *The XVI Nordic Congress of Wildlife Research*, 3–6.6.2016, Rovaniemi, Finland.
- Jones, B.C., Kleitch, J.L., Harper, C.A., Buehler, D.A., 2008. Ruffed grouse brood habitat use in a mixed hardwood forest: implications for forest management in the Appalachians. *For. Ecol. Manage.* 255 (10), 3580–3588. <http://dx.doi.org/10.1016/j.foreco.2008.02.019>.
- Kurki, S., Nikula, A., Helle, P., Lindén, H., 2000. Landscape fragmentation and forest composition effects on grouse breeding success in boreal forests. *Ecology* 81, 1985–1997.
- Kvasnes, M.A.J., Storaas, T., 2007. Effects of harvest regime on food availability and cover from predators in capercaillie (*Tetrao urogallus*) habitats. *Scand. J. Forest Res.* 22, 241–247.
- Lande, U., Herfindal, I., Willebrand, T., Moa, P.-F., Storaas, T., 2014. Landscape characteristics explain large-scale variation in demographic traits in forest grouse. *Landscape Ecol.* 29 (1), 127–139. <http://dx.doi.org/10.1007/s10980-013-9960-3>.
- Lindén, H., 1981. Estimation of juvenile mortality in the capercaillie, *Tetrao urogallus*, and the black grouse, *Tetrao tetrix*, from indirect evidence. *Finnish Game Res.* 39, 35–51.
- Lindén, H., Helle, E., Helle, P., Wikman, M., 1996. Wildlife triangle scheme in Finland: methods and aims for monitoring wildlife populations. *Finnish Game Res.* 49, 4–11.
- Lindén, H., Milonoff, M., Wikman, M., 1984. Sexual differences in growth strategies of capercaillie, *Tetrao urogallus*. *Finnish Game Res.* 42, 29–35.
- Lindén, H., Wikman, M., 1983. Goshawk predation on tetraonids: availability of prey and diet of the predator during breeding season. *J. Anim. Ecol.* 52, 953–968.
- Lindén, M., Rothsten-Lilja, S., Saaristo, L., Tokoi-Keto, P. (Eds.), 2014. Hyvän metsänhoidon suosituksheet – Riistametsänhoito. Metsätalouden kehittämisskeskus Tapion julkaisuja. (In Finnish only).
- Lone, K., van Beest, F.M., Mysterud, A., Gobakken, T., Milner, J.M., Ruud, H.-P., Loe, L. E., 2014. Improving broad scale forage mapping and habitat selection analyses with airborne laser scanning: the case of moose. *Ecosphere* 5 (11). <http://dx.doi.org/10.1890/ES14-00156.1>.
- Ludwig, G.X., 2007. Mechanisms of population declines in boreal forest grouse PhD thesis. University of Jyväskylä, Department of Biological and Environmental Science, Jyväskylä, Finland.
- Ludwig, G.X., Alatalo, R.V., Helle, P., Nissinen, K., Siitari, H., 2008. Large-scale drainage and breeding success in boreal forest grouse. *J. Appl. Ecol.* 45, 325–333. <http://dx.doi.org/10.1111/j.1365-2664.2007.01396.x>.
- Luke, 2015. Finnish National Forest Inventory Available at: <http://www.metla.fi/ohjelma/vmi/info-en.htm>. accessed October 12, 2015.
- MacArthur, R.H., MacArthur, J.W., 1961. On bird species diversity. *Ecology* 42 (3), 594–598.
- Melin, M., 2015. Studying Habitat Use and Behavior of Moose (*Alces alces*) by Integrating Airborne Laser Scanning and GPS Tracking. PhD Thesis. Dissertationes Forestales 209. <http://dx.doi.org/10.14214/df.2089>.
- Melin, M., Matala, J., Mehtälä, L., Pusenius, J., Packalen, P., 2016. Ecological dimensions of airborne laser scanning – analyzing the role of forest structure in moose habitat use within a year. *Remote Sens. Environ.* 173, 238–247. <http://dx.doi.org/10.1016/j.rse.2015.07.025>.
- Melin, M., Packalen, P., Matala, J., Mehtälä, L., Pusenius, J., 2013. Assessing and modeling moose (*Alces alces*) habitats with airborne laser scanning data. *Int. J. Appl. Earth Obs.* 23, 389–396.

- Miettinen, J., 2009. Capercaillie (*Tetrao urogallus*) Habitats in Managed Finnish Forests – the Current Status, Threats and Possibilities. *Dissertationes Forestales*, 90.
- Miettinen, J., Helle, P., Nikula, A., Niemelä, P., 2008. Large-scale landscape composition and capercaillie (*Tetrao urogallus*) density in Finland. *Ann. Zool. Fenn.* 45 (3), 161–173. <http://dx.doi.org/10.5735/086.045.0301>.
- Miettinen, J., Helle, P., Nikula, A., Niemelä, P., 2010. Capercaillie (*Tetrao urogallus*) habitat characteristics in north-boreal Finland. *Silva Fenn.* 44 (2), 235–254.
- Müller, J., Vierling, K., 2014. Assessing biodiversity by airborne laser scanning. In: Maltamo, M., Næsset, E., Vauhkonen, J. (Eds.), *Forestry Applications of Airborne Laser Scanning: Concepts and Case Studies*. Springer, Netherlands, Dordrecht, The Netherlands, pp. 357–374.
- Niemi, G., Hanowski, J., Helle, P., Howe, R., Mönkkönen, M., Venier, L., Welsh, D., 1998. Ecological sustainability of birds in boreal forests. *Conserv. Ecol.* 2 (2), 17.
- Noss, R.F., 1990. Indicators for monitoring biodiversity – a hierarchical approach. *Conserv. Biol.* 4 (4), 355–364.
- Palminteri, S., Powell, G.V.N., Asner, G.P., Peres, C.A., 2012. LiDAR measurements of canopy structure predict spatial distribution of a tropical mature forest primate. *Remote Sens. Environ.* 127, 98–105.
- Pakkala, T., Pellikka, J., Linden, H., 2003. Capercaillie *Tetrao urogallus* – a good candidate for an umbrella species in taiga forests. *Wildlife Biol.* 9, 309–316.
- Pellikka, J., Rita, H., Lindén, H., 2005. Monitoring wildlife richness – Finnish applications based on the wildlife triangle censuses. *Ann. Zoo. Fenn.* 42, 123–134.
- Picozzi, N., Hepburn, L.V., 1986. A study of black grouse in north-east Scotland. In: *Proceedings of the 3rd International Grouse Symposium*, pp. 462–480.
- R Core Team, 2015. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. URL <<https://www.R-project.org/>>.
- Rhim, S.-J., Son, S.-H., Hwang, H.-S., 2015. Factors affecting chick mortality of hazel grouse in a temperate forest, South Korea. *Forest. Ecol. Manage.* 348, 92–96.
- Shepard, D., 1968. A two-dimensional interpolation function for irregularly-spaced data. In: *Proceedings of the 1968 ACM National Conference*, New York, pp. 517–524.
- Sirkiä, S., Lindén, A., Helle, P., Nikula, A., Knappe, J., Lindén, H., 2010. Are the declining trends in forest grouse populations due to changes in the forest age structure? A case study of Capercaillie in Finland. *Biol. Conserv.* 143, 1540–1548.
- Sjöberg, K., 1996. Modern forestry and the capercaillie. In: DeGraaf, M., Miller, R.I. (Eds.), *Forested Landscapes*. Chapman & Hall, London, pp. 111–135.
- Storch, I., 1993. Habitat selection by capercaillie in summer and autumn: Is bilberry important? *Oecologia* 95, 257–265.
- Swenson, J.E., 1991. Social Organization of Hazel Grouse and Ecological Factors Influencing It PhD Thesis. Univ. Alberta, Edmonton, Canada, p. 185.
- Swenson, J.E., Angelstam, P., 1993. Habitat separation by sympatric forest grouse in Fennoscandia in relation to boreal forest succession. *Can. J. Zool.* 71 (7), 1303–1310.
- Vauhkonen, J., Imponen, J., 2016. Unsupervised classification of airborne laser scanning data to locate potential wildlife habitats for forest management planning. *Forestry*. <http://dx.doi.org/10.1093/forestry/cpw011>.
- Vierling, K.T., Vierling, L.A., Gould, W.A., Martinuzzi, S., Clawges, R.M., 2008. Lidar: shedding new light on habitat characterization and modeling. *Front. Ecol. Environ.* 6 (2), 90–98.
- Wegge, P., Olstad, T., Gregersen, H., Hjeljord, O., Sivkov, A.V., 2005. Capercaillie broods in pristine boreal forest in northwestern Russia: the importance of insects and cover in habitat selection. *Can. J. Zool.* 83, 1547–1555.
- Weidong, G., Swihart, R.K., 2004. Absent or undetected? Effects of non-detection of species occurrence on wildlife-habitat models. *Biol. Conserv.* 116 (2), 195–203. [http://dx.doi.org/10.1016/S0006-3207\(03\)00190-3](http://dx.doi.org/10.1016/S0006-3207(03)00190-3).
- Zellweger, F., Braunisch, V., Baltensweiler, A., Bollmann, K., 2013. Remotely sensed forest structural complexity predicts multi species occurrence at the landscape scale. *Forest. Ecol. Manage.* 307, 303–312.
- Zellweger, F., Baltensweiler, A., Ginzler, C., Roth, T., Braunisch, V., Brugmann, H., Bollmann, K., 2016. Environmental predictors of species richness in forest landscapes: abiotic factors versus vegetation structure. *J. Biogeogr.* <http://dx.doi.org/10.1111/jbi.12696>.